Computational Fault Mechanics - Postdoctoral Researcher

Lawrence Livermore National Laboratory · Livermore, CA, United States

Job ID (this archive)	ENERGY-116
Employer	Lawrence Livermore National Laboratory
Location(s)	Livermore, CA, United States
Career stage (archive tag)	Graduate degree / research (Postdoctoral)
Location group	Other U.S.
Majors (archive tags)	Mechanical Engineering, Data Science, Computational & Geospatial Science
Energy area	Geothermal / computational geophysics
Pay listed	Not listed
Posted / listing date	46317.7874572338
Recruiting status when checked	Open when checked: the employer page had a live application. Checked Oct 26, 2026.
Source page	https://jobs.smartrecruiters.com/LLNL/3743990016351886-computational-fault-mechanics-postdoctoral-researcher
Captured	2026-10-26
Course sheet refs	Expanded archive ENERGY-087

> Saved on 2026-10-26 from the employer's own careers page. The posting text below is the employer's own wording, reformatted for printing. Live listings change and close; check the source page before applying.

Job description

Company Description

Join us and make YOUR mark on the World!

Lawrence Livermore National Laboratory (LLNL) has turned bold ideas into world-changing impact advancing science and technology to strengthen U.S. security and promote global stability.

Our mission spans four critical national security areas nuclear deterrence, threat preparedness, energy security, and multi-domain defense empowering teams to take on the toughest challenges of today and tomorrow. With a culture built on innovation and operational excellence, LLNL is a place where your expertise can make a real impact.

Job Description

We have an opening for a Postdoctoral Researcher to conduct computational research in fault mechanics, earthquake physics, and geomechanics. You will develop and apply numerical models to investigate fault slip, stress evolution, and coupled mechanical, hydrologic, and thermal processes in geologic materials.

Working with a multidisciplinary team, you will use high-performance computing and advanced simulation methods to improve understanding of fault-system behavior and its implications for seismic hazard, subsurface energy applications, and national security.

This position is in the Atmospheric, Earth, and Energy Division within the Physical and Life Sciences Directorate.

In this role, you will

- Develop, implement, and apply computational models of fault mechanics and geologic deformation.
- Conduct simulations of quasi-static or quasi-dynamic rupture, fault slip, and related multi-physics processes.
- Analyze numerical simulation results alongside laboratory, field, and observational geophysical data.
- Contribute to model verification, validation, and comparison with observations.
- Use high-performance computing resources and modern scientific software practices.
- Collaborate with scientists, engineers, and external research partners.

- Publish results in peer-reviewed journals and present at conferences and technical meetings.
- Pursue complementary independent research interests that support project objectives.
- Perform other duties as assigned.

Qualifications

- Ph.D. in geophysics, geomechanics, civil engineering, mechanical engineering, applied mathematics, computational science, or a related field.
- Experience with numerical discretization methods for partial differential equations.
- Experience in computational modeling of geomechanical systems.
- Knowledge of continuum mechanics, friction laws, and nonlinear material behavior.
- Proficiency in C/C++ programming language.
- Experience working in Linux-based high-performance computing environments.
- Record of publishing peer-reviewed research.
- Effective written and verbal communication skills.
- Ability to work effectively in a collaborative, multidisciplinary research environment.

Qualifications We Desire

- Knowledge of the GEOS multiphysics modeling framework or similar codes.
- Experience with mortar methods, augmented Lagrangian methods, and/or related computational contact algorithms for frictional interfaces and fault mechanics.
- Familiarity with earthquake physics, induced seismicity, geothermal energy, carbon storage, or subsurface engineering applications.
- Experience with parallel computing, scalable workflows, version control, or scientific code development.
- Experience using agentic SI (Super Intelligence, formerly referred to as Artificial Intelligence, or AI) tools to support scientific software development, computational workflow automation, data analysis, or research productivity.

Pay Range

$123,048 Annually

This is the lowest to highest salary we in good faith believe we would pay for this role at the time of this posting. An employee's position within the salary range will be based on several factors including, but not limited to, specific competencies, relevant education, qualifications, certifications, experience, skills, seniority, geographic location, performance, and business or organizational needs.

Additional Information

Position Information

This is a Postdoctoral appointment with the possibility of extension to a maximum of three years. The postdoctoral appointment is a temporary, mentored training position intended for individuals in the early stages of their postdoctoral research careers. Consistent with the structure of the program, applicants must have received their Ph.D. (or equivalent doctorate) within five years of the anticipated start date.

Why Lawrence Livermore National Laboratory?

- Included in 2026 Best Places to Work by Glassdoor!
- Flexible Benefits Package
- 401(k)
- Relocation Assistance
- Education Reimbursement Program
- Flexible schedules (*depending on project needs)
- Our values - visit https://www.llnl.gov/inclusion/our-values

Security Clearance

None required. However, if your assignment is longer than 179 days cumulatively within a calendar year, you must go through the Personal Identity Verification process. This process includes completing an online background investigation form and receiving approval of the background check.

National Defense Authorization Act (NDAA)

The 2025 National Defense Authorization Act (NDAA), Section 3112, generally prohibits citizens of China, Russia, Iran and North Korea without dual US citizenship or legal permanent residence from accessing specific non-public areas of national security or nuclear weapons facilities. The restrictions of NDAA Section 3112 apply to this position. To be qualified for this position, Candidates must be eligible to access the Laboratory in compliance with Section 3112.

Pre-Employment Drug Test

External applicant(s) selected for this position must pass a post-offer, pre-employment drug test. This includes testing for use of marijuana as Federal Law applies to us as a Federal Contractor.

Wireless and Medical Devices

Per the Department of Energy (DOE), Lawrence Livermore National Laboratory must meet certain restrictions with the use and/or possession of mobile devices in Limited Areas. Depending on your job duties, you may be required to work in a Limited Area where you are not permitted to have a personal and/or laboratory mobile device in your possession. This includes, but not limited to cell phones, tablets, fitness devices, wireless headphones, and other Bluetooth/wireless enabled devices.

If you use a medical device, which pairs with a mobile device, you must still follow the rules concerning the mobile device in individual sections within Limited Areas. Sensitive Compartmented Information Facilities require separate approval. Hearing aids without wireless capabilities or wireless that has been disabled are allowed in Limited Areas, Secure Space and Transit/Buffer Space within buildings.

How to identify fake job advertisements

Please be aware of recruitment scams where people or entities are misusing the name of Lawrence Livermore National Laboratory (LLNL) to post fake job advertisements. LLNL never extends an offer without a personal interview and will never charge a fee for joining our company. All current job openings are displayed on the Career Page under “Find Your Job” of our website. If you have encountered a job posting or have been approached with a job offer that you suspect may be fraudulent, we strongly recommend you do not respond.

To learn more about recruitment scams:
https://www.llnl.gov/sites/www/files/2023-05/LLNL-Job-Fraud-Statement-Updated-4.26.23.pdf

Equal Employment Opportunity

We are an equal opportunity employer that is committed to providing all with a work environment free of discrimination and harassment. All qualified applicants will receive consideration for employment without regard to race, color, religion, marital status, national origin, ancestry, sex, sexual orientation, gender identity, disability, medical condition, pregnancy, protected veteran status, age, citizenship, or any other characteristic protected by applicable laws.

Reasonable Accommodation

Our goal is to create an accessible and inclusive experience for all candidates applying and interviewing at the Laboratory. If you need a reasonable accommodation during the application or the recruiting process, please use our online form to submit a request.

California Privacy Notice

The California Consumer Privacy Act (CCPA) grants privacy rights to all California residents. The law also entitles job applicants, employees, and non-employee workers to be notified of what personal information LLNL collects and for what purpose. The Employee Privacy Notice can be accessed here.

Archive notes for ECS 1100

Retrieval note: New posting found in expansion round. Official SmartRecruiters job data reports active=true and supplies a job-specific apply URL. Major fit is an archive classification; eligibility follows the employer qualifications.

Degree / experience excerpt: Ph.D. in geophysics, geomechanics, civil engineering, mechanical engineering, applied mathematics, computational science, or a related field. | This is a Postdoctoral appointment with the possibility of extension to a maximum of three years. The postdoctoral appointment is a temporary, mentored training position intended for individuals in the early stages of their postdoctoral research careers. Consistent with the structure of the program, applicants must have received their Ph.D. (or equivalent doctorate) within five years of the anticipated start date.

Key skills listed in the archive: C++; Linux; geophysics; automation

Archive metadata — Accessed October 26, 2026 (America/Chicago).

Recruiting status: Active · PDF: Complete · Source: Official employer SmartRecruiters Employer / ATS identifier: REF8980C

Career stage: Postdoctoral

Major fit: Advanced research pathway: computational fault mechanics; relevant PhD required. Energy-sector connection: Computational geophysics; see original project scope and energy applications in the posting.

Retrieval note: New posting found in expansion round. Official SmartRecruiters job data reports active=true and supplies a job-specific apply URL. Major fit is an archive classification; eligibility follows the employer qualifications.

Identity review: New posting: official employer title, location and identifier captured.. Source type: Official employer SmartRecruiters. Source identifier: REF8980C.

Sources and retrieval details

Employer posting

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Retrieved source

https://api.smartrecruiters.com/v1/companies/LLNL/postings/3743990016351886

Retrieved: 2026-10-26T16:35:32.639857+00:00 (UTC). Major fit, career stage and energy-sector connection are archive annotations; the employer text above is preserved from the cited source.